MOVING-WINDOW ENGEL HILBERT COCYCLES WITH $\Omega(\sqrt{k})$ HOMOGENEOUS GAUGE DISTORTION

ARGUS OPEN RESEARCH PROJECT

Abstract. Toward the Hilbert-space Lang–Plaut bi-Lipschitz embedding problem for doubling metric spaces, we construct a family of moving-window injective Hilbert-space cocycles on finite balls of the Engel group that are uniformly doubling, admit an exact logarithmic gauge comparison, and incur distortion at least $c\sqrt{k}$ relative to any fixed homogeneous Engel gauge.

1. Main Construction Theorem

Theorem 1.1. *On the three-step nilpotent Engel group $\mathcal{E}$, there exists a sequence of moving-difference-window maps $F_k : B_{\mathcal{E}}(e, R_k) \to \ell_2$ such that:*

(1) *The induced metric balls are uniformly doubling with doubling constant independent of k.*
(2) *Relative to the homogeneous Carnot–Carathéodory gauge d_{CC}, the compression/distortion of F_k on $B_{\mathcal{E}}(e, R_k)$ is at least $c\sqrt{k}$ for an explicit absolute constant $c > 0$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000582 in the domain of metric geometry and bi-Lipschitz embeddings follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.
(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped construction theorem* deciding the scope: uniformly doubling finite-ball injective Hilbert cocycles on the Engel group with >= c*sqrt(k) distortion against a homogeneous Engel gauge.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unrestricted non-homogeneous Hilbert-space Lang-Plaut embedding problem.

References

[1] Primary problem list and source literature for Result 20000582 (metric geometry and bi-Lipschitz embeddings), archived with exact theorem locators and verification certificates in the accompanying repository package.

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